

IMCA 221
Programming for Artists
Fall 2025

lee wilkins
l.wilkins@concordia.ca

**Class is on Zoom even in the classroom
for sharing, find the details on Moodle**

<https://moodle.concordia.ca/>

Download the slides!



<https://macm.org/en/exhibitions/rafael-lozano-hemmer-unstable-presence/> Raphael Lozano Hemmer

<https://www.lozano-hemmer.com/images.php>



Daily Tous
Les Jours
[https://
www.dailytou
slesjours.com
/en/work](https://www.dailytouslesjours.com/en/work)



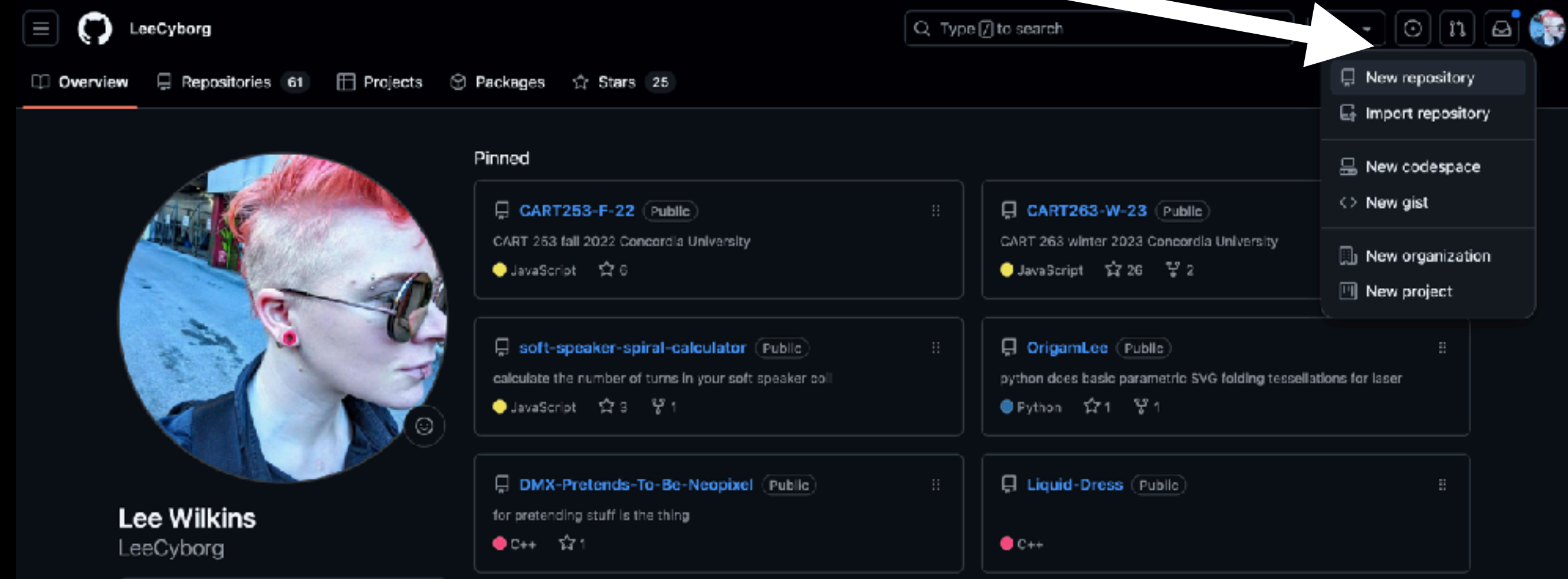
**Mapp
Montreal
Festival
[https://
www.mappmt
l.com/en](https://www.mappmtl.com/en)**

GitHub

Github is a place you can store and edit code. It is version control so you can go back and look at old code or collaborate with others.

Our class has a github!

Repositories are like projects that hold code. Create a new repository for each project.



Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

Owner *

LeeCyborg

Repository name *

Great repository names are short and memorable. Need inspiration? How about [glowing-disco](#) ?

Description (optional)

☒ Public

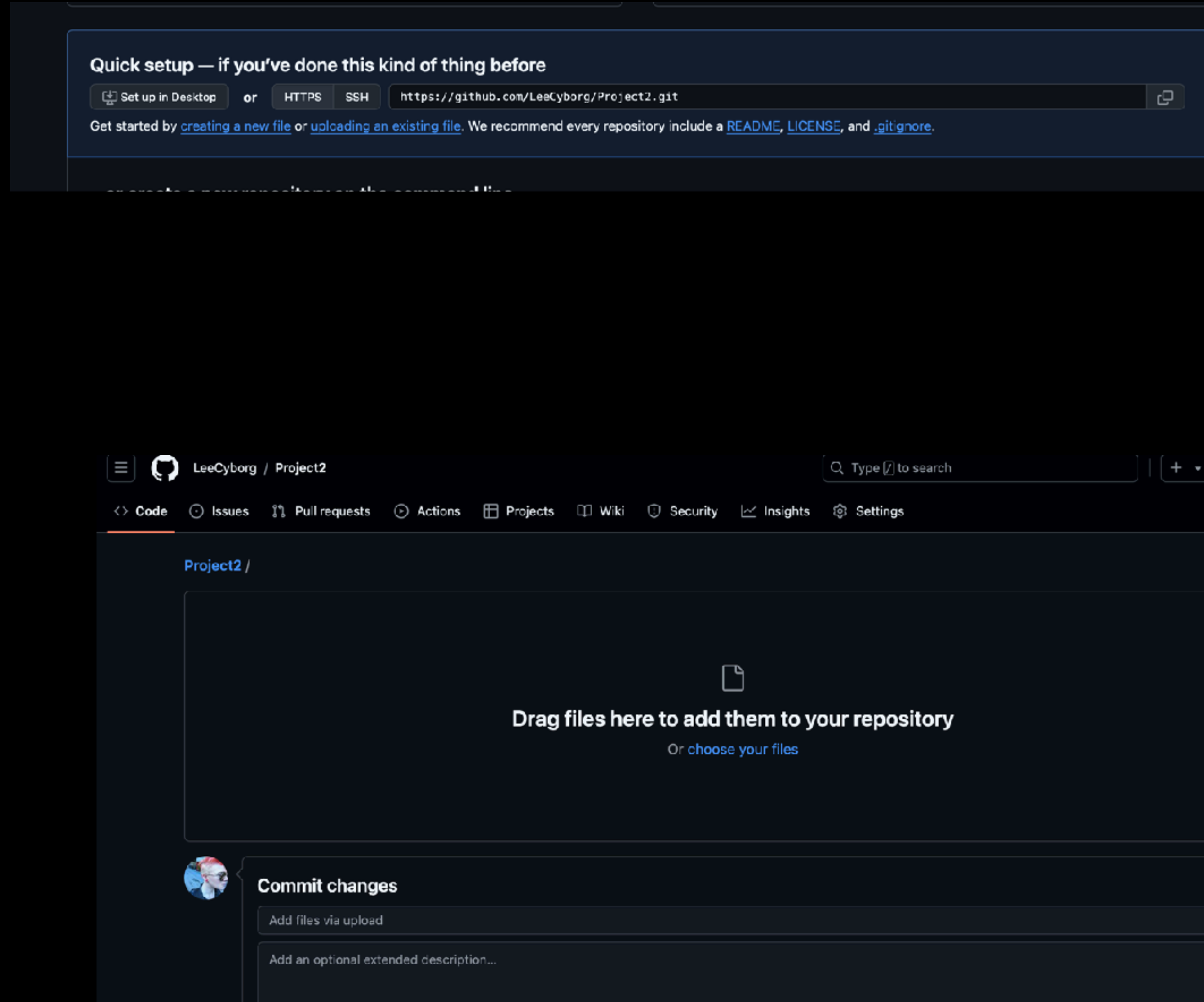
Anyone on the internet can see this repository. You choose who can commit.

☐ Private

You choose who can see and commit to this repository.

GitHub.com

You can use a variety of ways to add files to your repository. For now, we can just use drag and drop in the browser. But you can use GitHub Desktop or the command line if you want.



GitHub

Submit your repository link for your project (if you want). Create a new repository of each project.

Resources:

Max Documentation: <https://docs.cycling74.com/max8>

Max Cookbook <https://music.arts.uci.edu/dobrian/maxcookbook/>

Andrew Robinson Video Tutorials <https://www.youtube.com/@AndrewRobinson26>

Keyboard Shortcuts

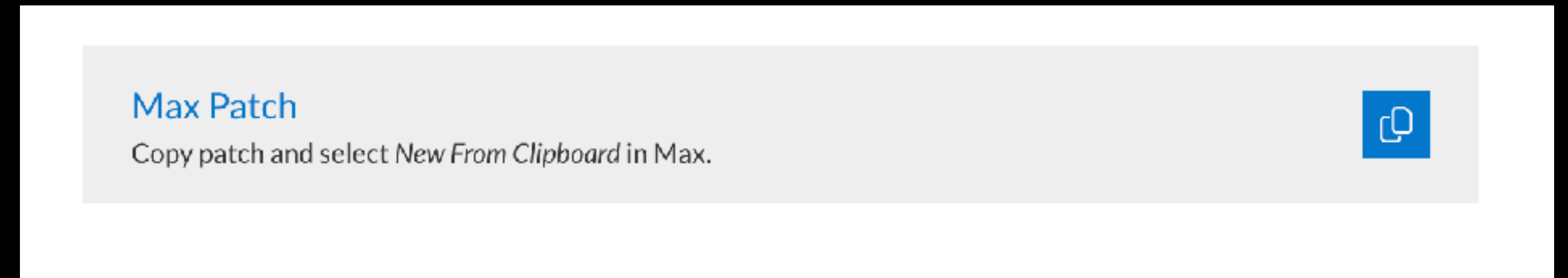
<https://docs.cycling74.com/max8/vignettes/keycommands>

- a: attrui.
- b: button.
- c: comment.
- f: floating point number box.
- h: briefly highlights a small area around the cursor.
- H: A capital letter "H" briefly highlights a larger area around the cursor.
- i: integer number box.
- j: object box containing "jit." for creating Jitter objects.
- l: object box containing "live." for creating Live objects.
- m: message.
- n: new blank object with the cursor active. Typing the name of any object and pressing enter or clicking outside of the object box will transform it into that object.
- r: bring up a list of the most recently created objects, including any arguments and attributes typed in. Selecting an element from the list creates an object with the corresponding text and with the cursor active at the far right. Hitting a carriage return or clicking away from the object instantiates the object.
- p: create an object box containing the message newobj @presentation 1 @text and a cursor. When you type the name of an object (e.g. dial) and hit a carriage return, the object will transform itself into a copy of the object whose name you type in, and the object will be automatically added to the Presentation Layer.
- s: slider.
- t: toggle.
- x: shows a menu describing the key commands, including those added by external packages.
- z: zooms the patcher in around the cursor.
- Z: A capital letter "Z" zooms the patcher out around the cursor.

You can copy Max code and paste it directly into your patch as text.

CMD + V (paste)

```
{
  "boxes" : [ {
    "box" : {
      "maxclass" : "gain~",
      "patching_rect" :
[ 385.792366743087769,
357.923513054847717, 22.0, 140.0 ],
      "outlettype" : [ "signal", "" ],
      "multichannelvariant" : 0,
      "id" : "obj-114",
      "parameter_enable" : 0,
      "numinlets" : 1,
      "numoutlets" : 0
    }
  ]
}
```



Video Experiment next week (midnight, after class)

- Explore audio tools we worked on in class
- Practice and explore tutorials online
- Create an experiment, which could be the beginning of a future project
- Your experiment should have a concept, even though it is not fully developed.

To hand in your project, use the project template on Moodle saved file (File > Save as Project) .maxpat

- A PDF that contains project documentation. See Project_Documentation_template
- A clear, strong image of your project.
- A screen capture of your max patch
- A link to video or audio recording of your project working (can be a video or a link to a private video on YouTube, Vimeo or Google)
- A 50-100 word explanation of your project inside your maxpatch Be sure to name files properly (no untitled-1.zip) All files are expected to be cleaned up and arranged in a reasonable, legible way. Videos should be clear, well light and show your project working.

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My Project Title

Lee Wilkins



Main Image:

Include a clear, strong image of your work. If the work is just audio, include an image of whatever context you imagine it in.

Artist Statement:

Write an artist statement that describes your work. Describe the work as though it is on a gallery wall, what would you read? 50-100 words. Remember, even a small project should have some kind of theme or concept. What do you want people to experience while viewing your work?

A good place to start is:

[Project Name] is an exploration of [concept]. By using [something, audio, video, samples from something?] viewers are able to experience [what do you want them to feel?].

Project Template on Moodle has everything you need, just replace the text and images with your own.

Video Experiment next week (midnight, after class)

25% Functionality / technical

25% creativity / concept

25% execution and quality

25% documentation

My Project Title

Lee Wilkins



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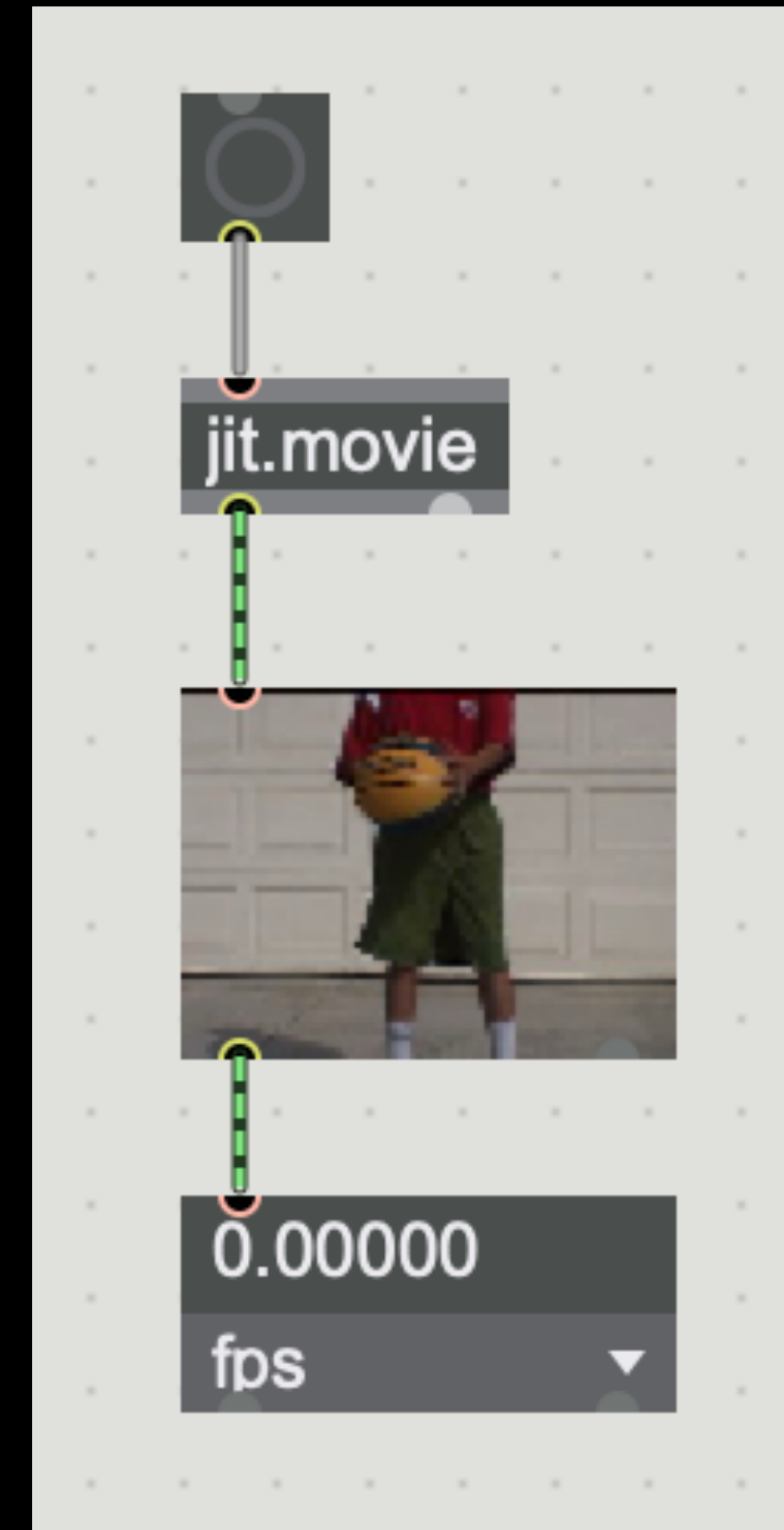
[Project Name] is an exploration of [concept]. By using [something, audio, video, samples from something?] viewers are able to experience [what do you want them to feel?].

Project Template on Moodle has everything you need, just replace the text and images with your own.

Jitter is the video/
visual aspect of
maxMSP

All jitter objects
start with jit.

Jitter objects
usually have green
or blue patch cords.



Make a view port by making a `jit.world` object, which creates an external window. Or `jit.pworld` which creates a window inside your patch.

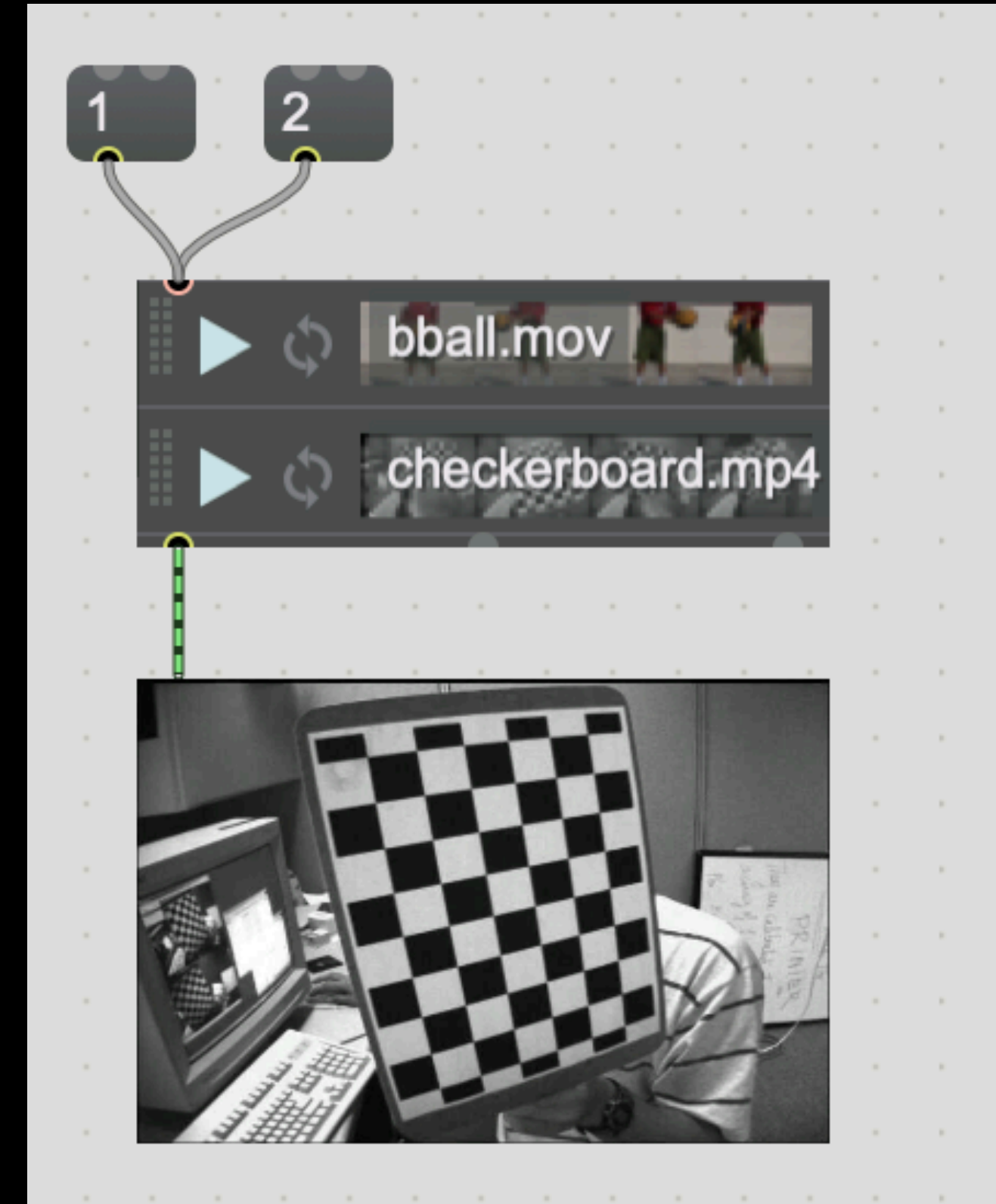
We can see a sub-set of those features by using `jit.pwindow`



Jitter has an object similar to playlist. Called `jit.playlist`.

Drag and drop videos similar to sound files, connect them to the `pworld` or `pwindow`

Notice the green patch cords, they are from jitter matrices!



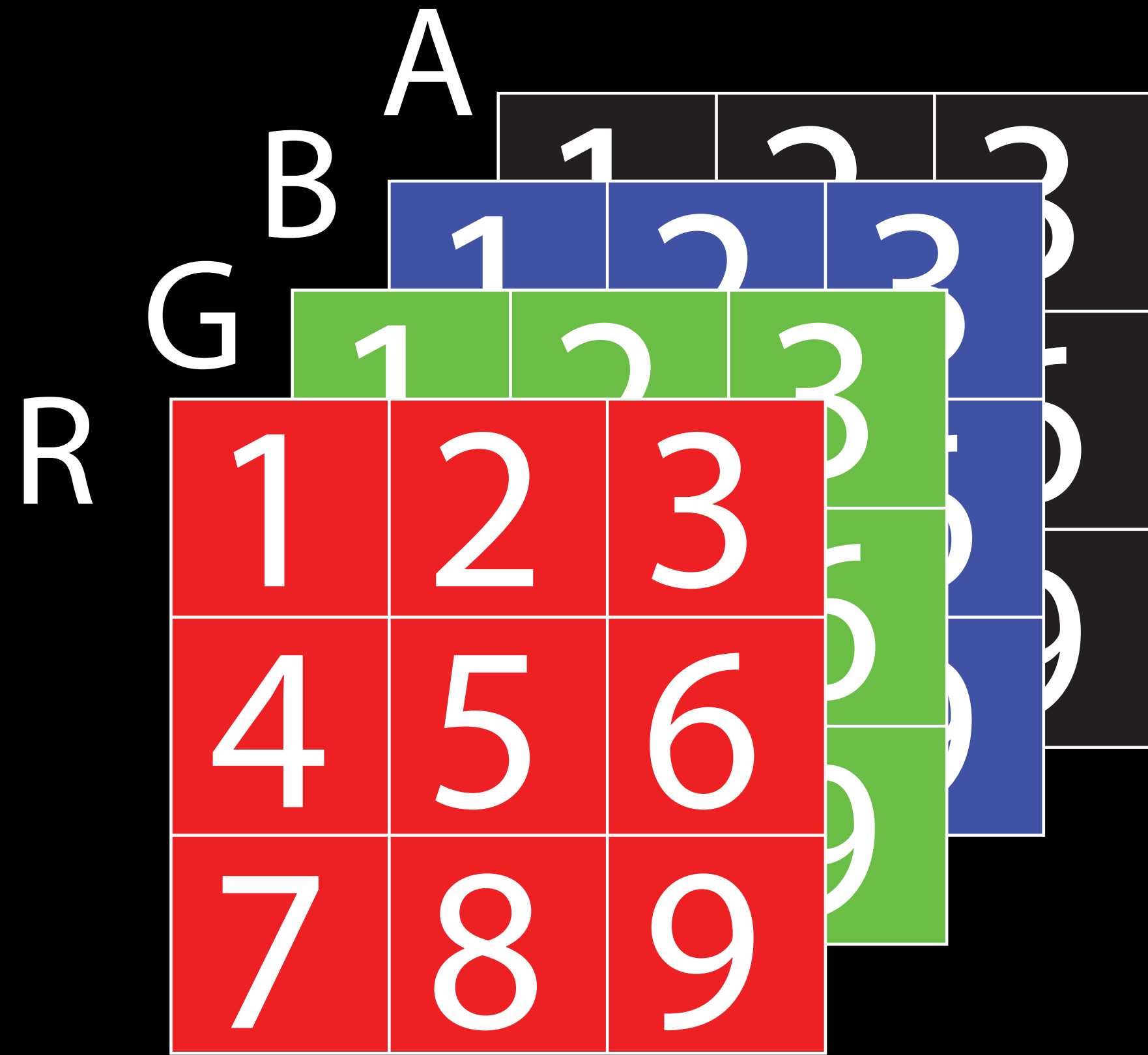
Matrix

Jitter uses a matrix to display graphics. Matrices are grids with cells. Each cell is labeled like this. It is not like a cartesian plane.

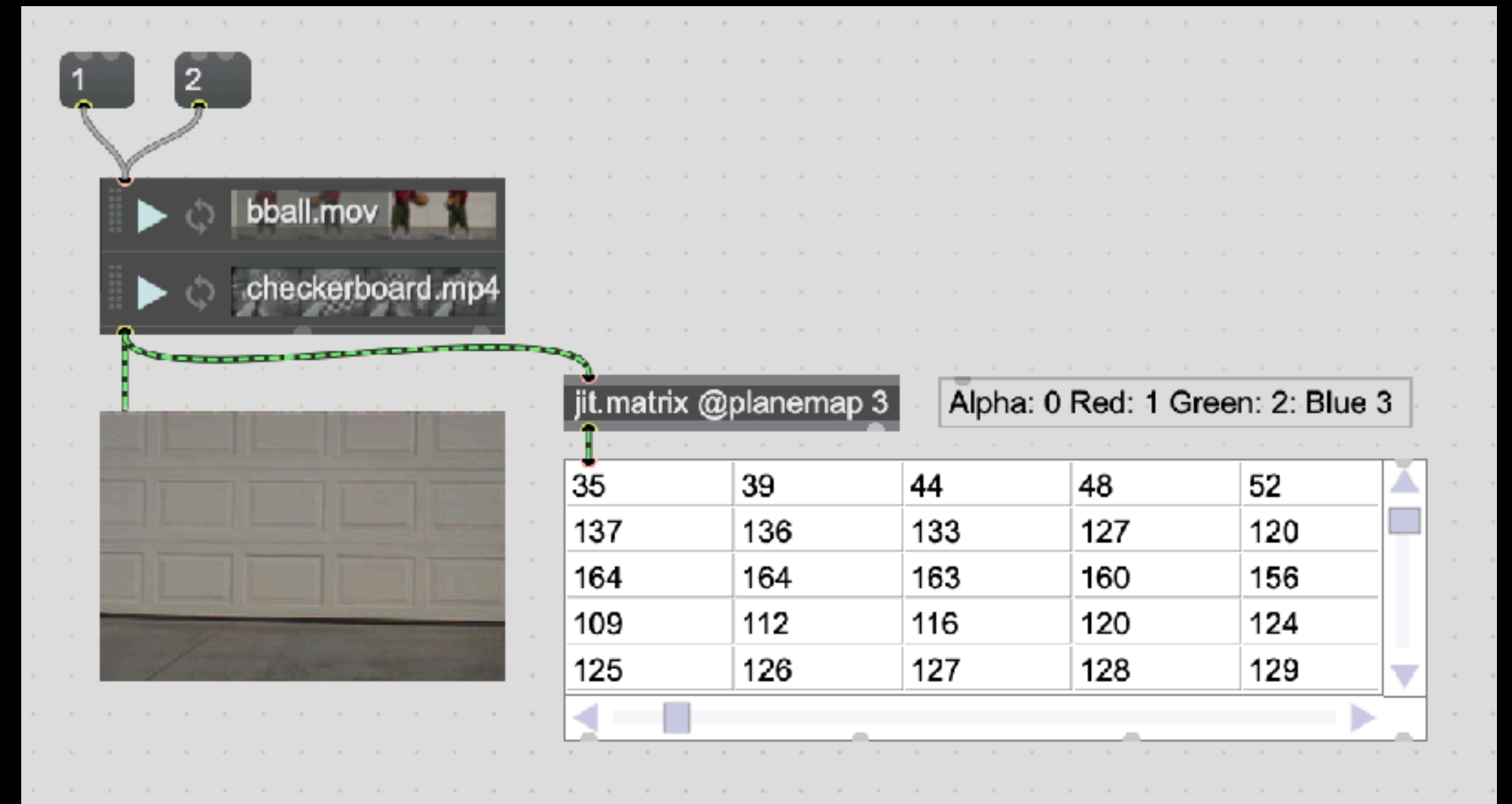
1	2	3
4	5	6
7	8	9

Matrix

Each matrix can have multiple planes. Planes are basically maps that describe each pixel. There are 4 here: RGBA, Red, Green, Blue, Alpha. Alpha is transeparcy. They are



We can view this matrix using `jit.cellblock`. Add plane map attribute to see each plane. This jsut lets us understand what exactly a matrix is.



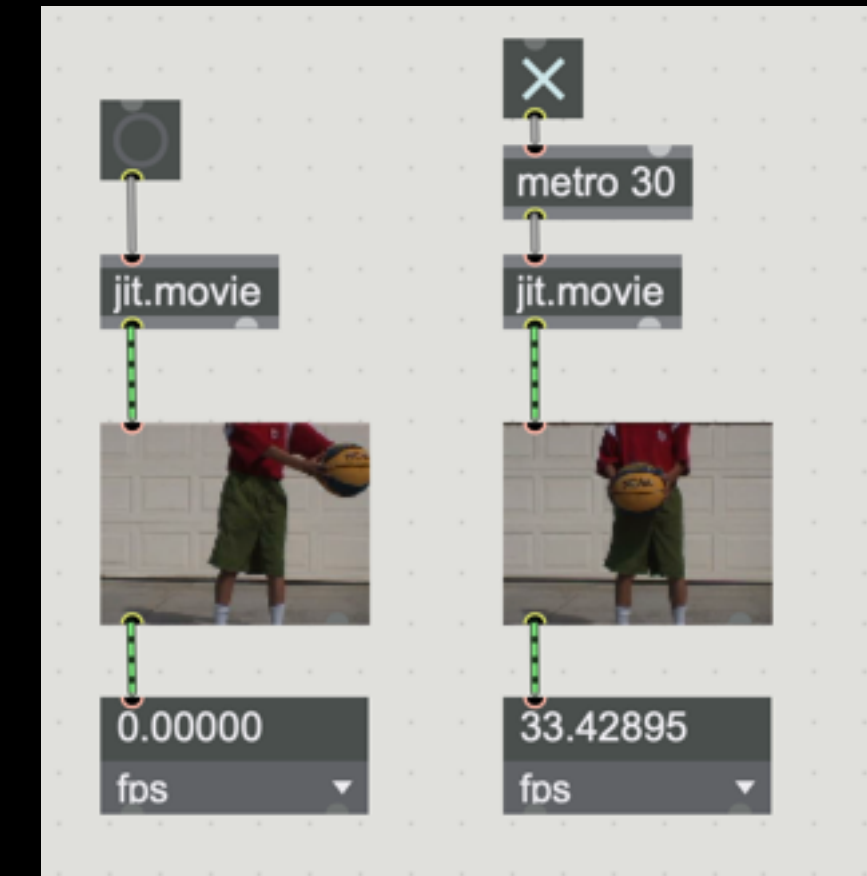
Matricies can be
manipulated
manually by using
attrui



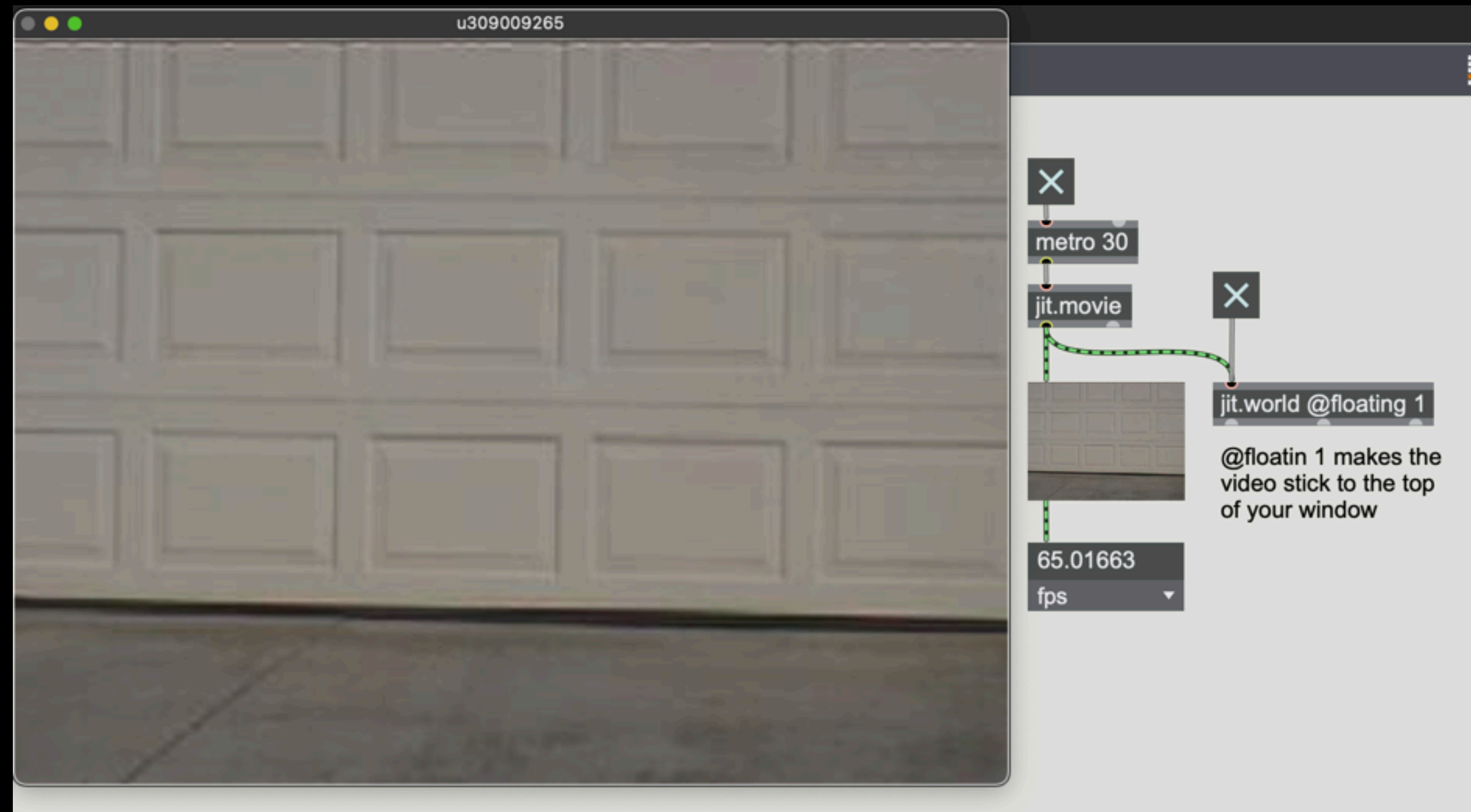
We can also use more powerful objects to play movies, such as `jit.movie`.

Try with a button, a single frame advances. With a `metro` we can keep triggering it. Metro 30 is around 30 FPS (give or take).

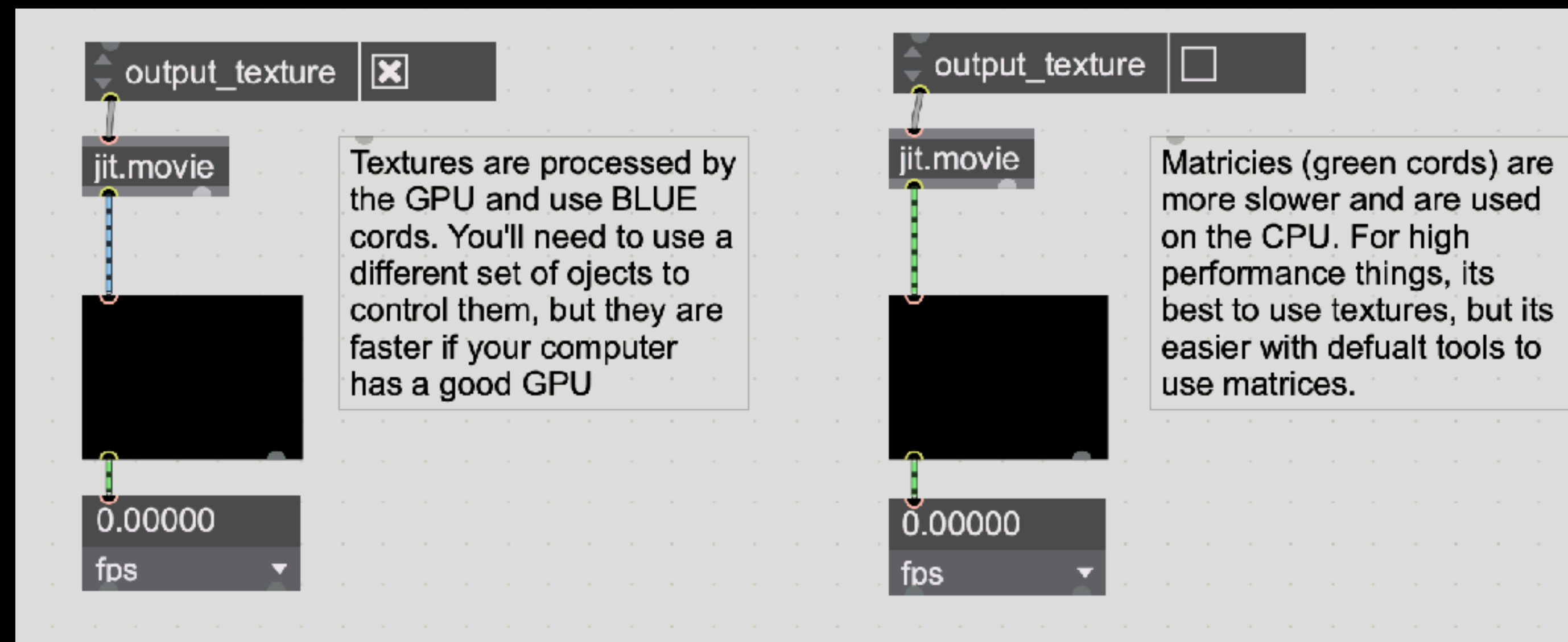
View this with `jit.fps` object



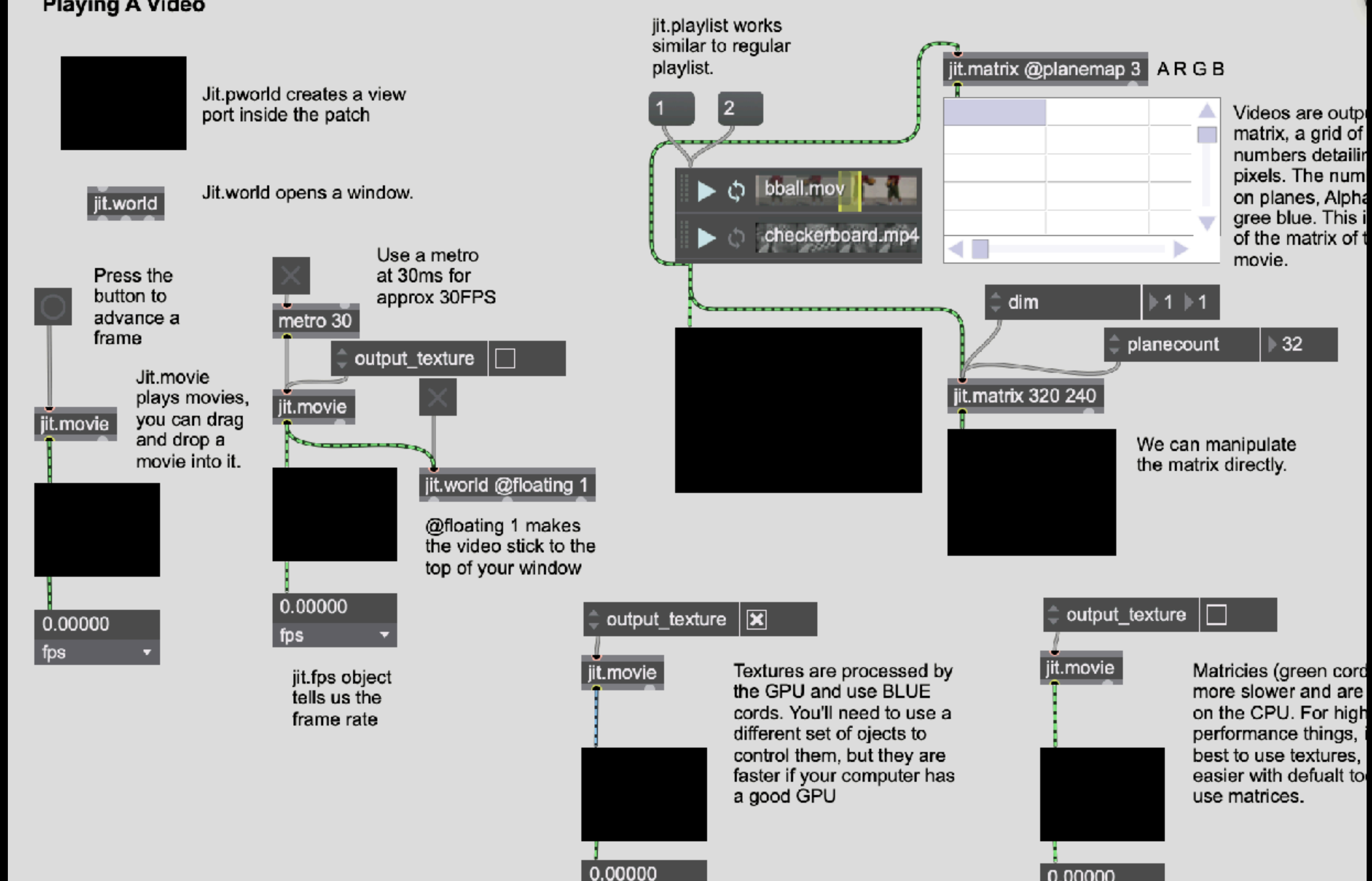
Use `jit.world` to
make a window.
`@floating 1` makes
the window stick
to the top.



Jitter can use matrices OR textures to play movies. Each has pros and cons. Textures use your GPU, matrices use your CPU. They are each used differently. By default, many effects are used with matrices.



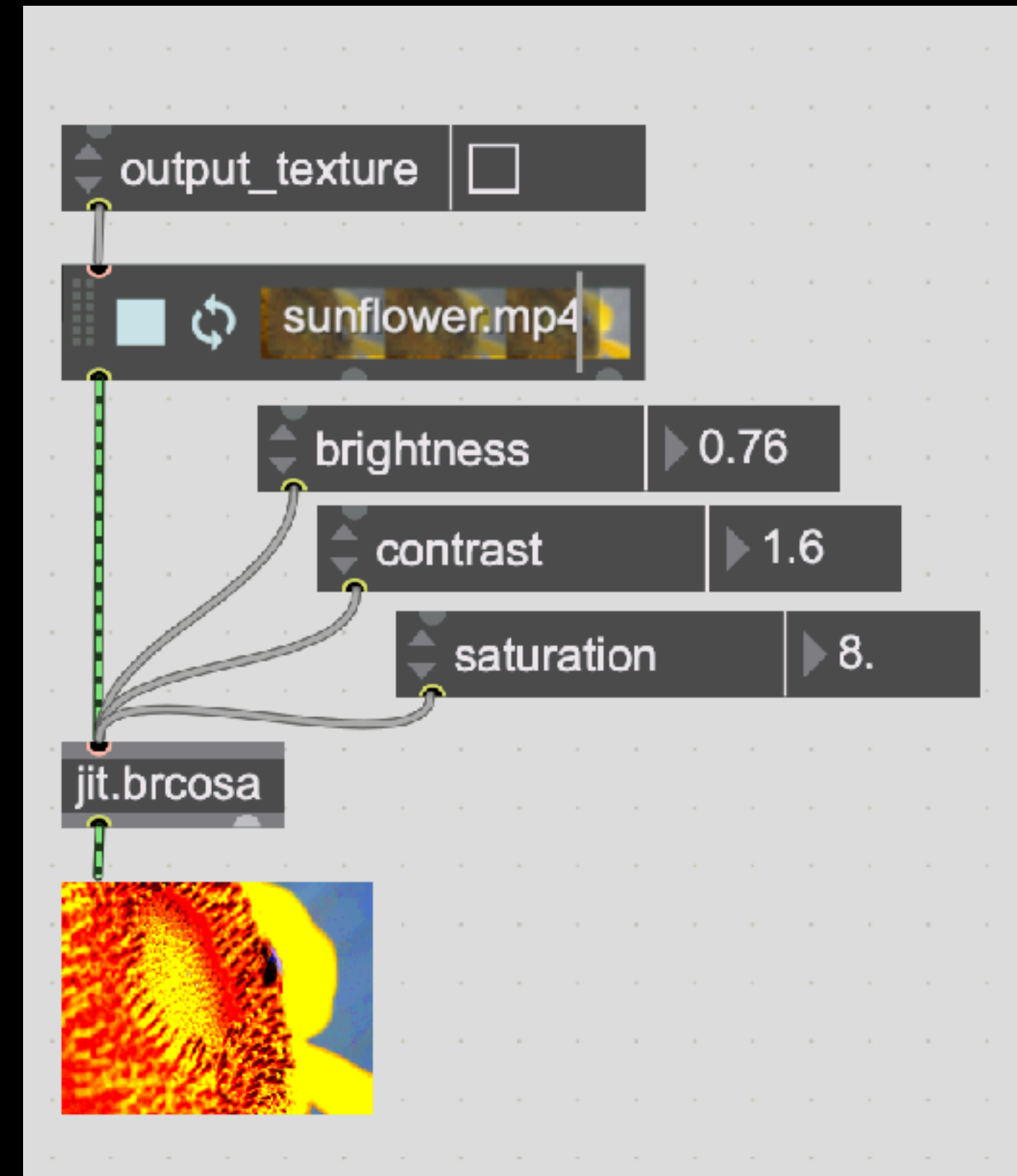
Playing A Video



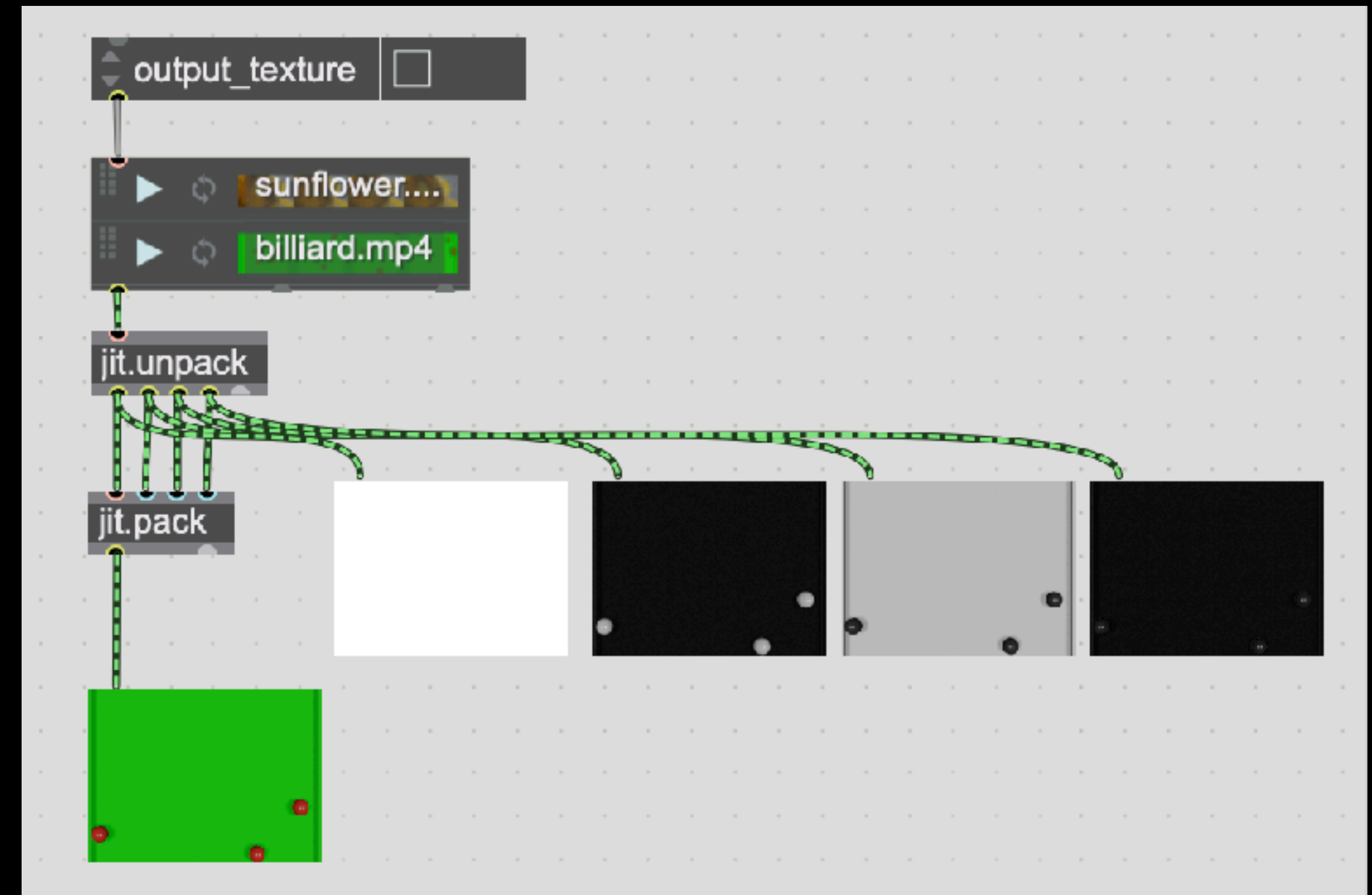
Playing a video maxpatch

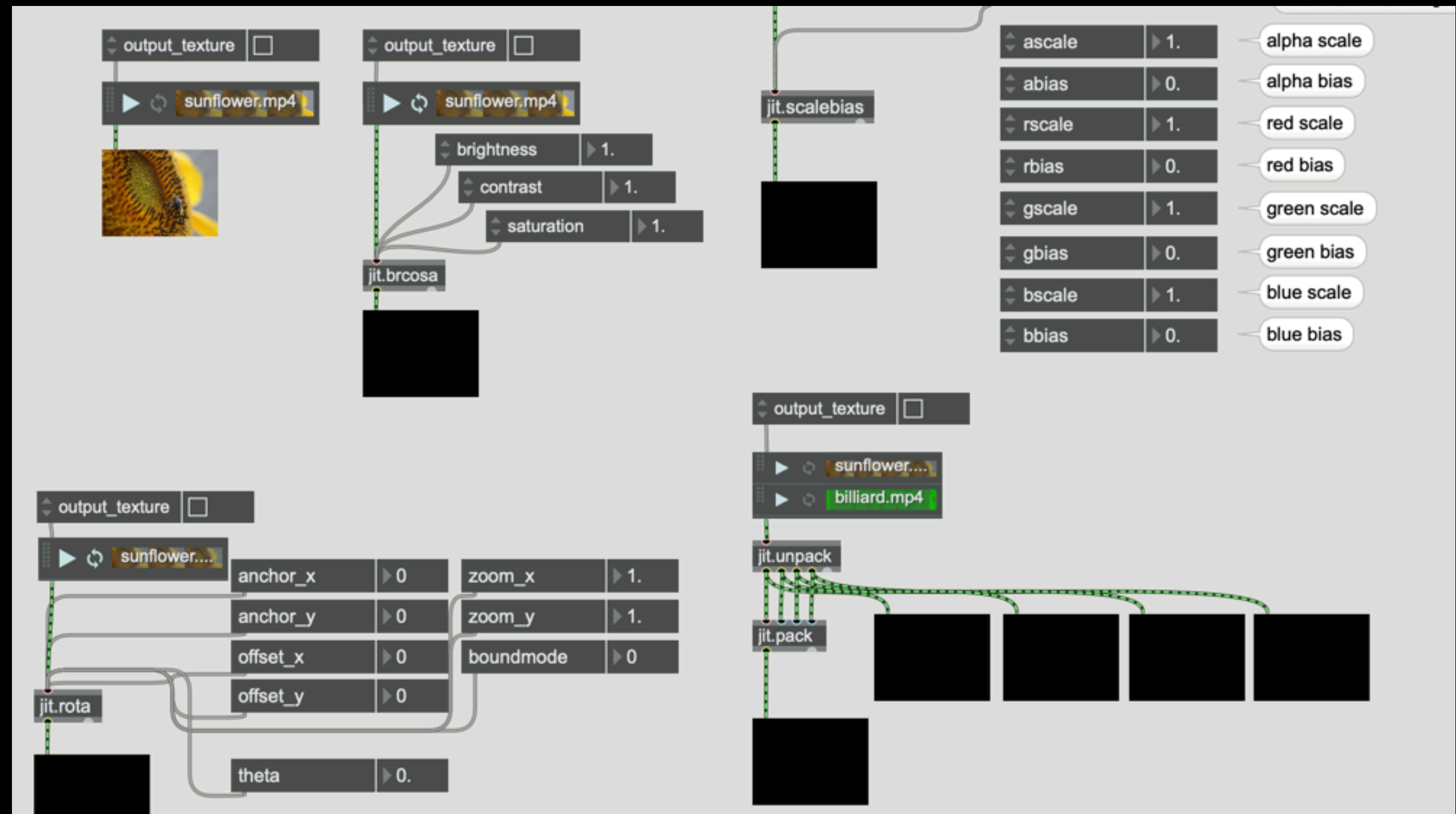
Use the attrui object to ensure **output_texture** is **NOT** checked in order to use default processing tools.

jit.brcosa is a good tool to change brightness, contrast, and saturation.



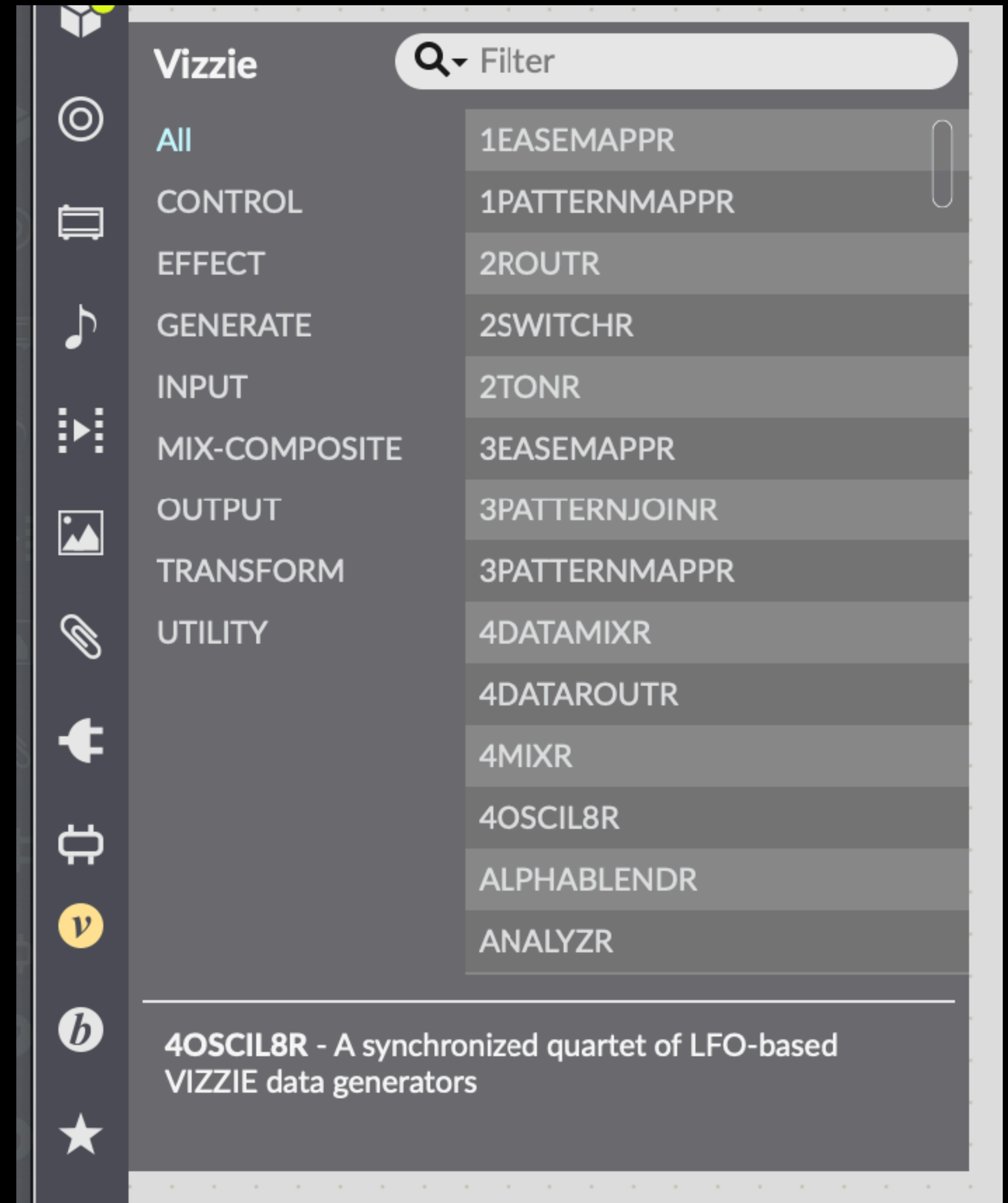
You can unpack each layer and manipulate it individually using `jit.unpack`, then `jit.pack`. Try using `jit.rota` on each matrix layer and then repacking them.



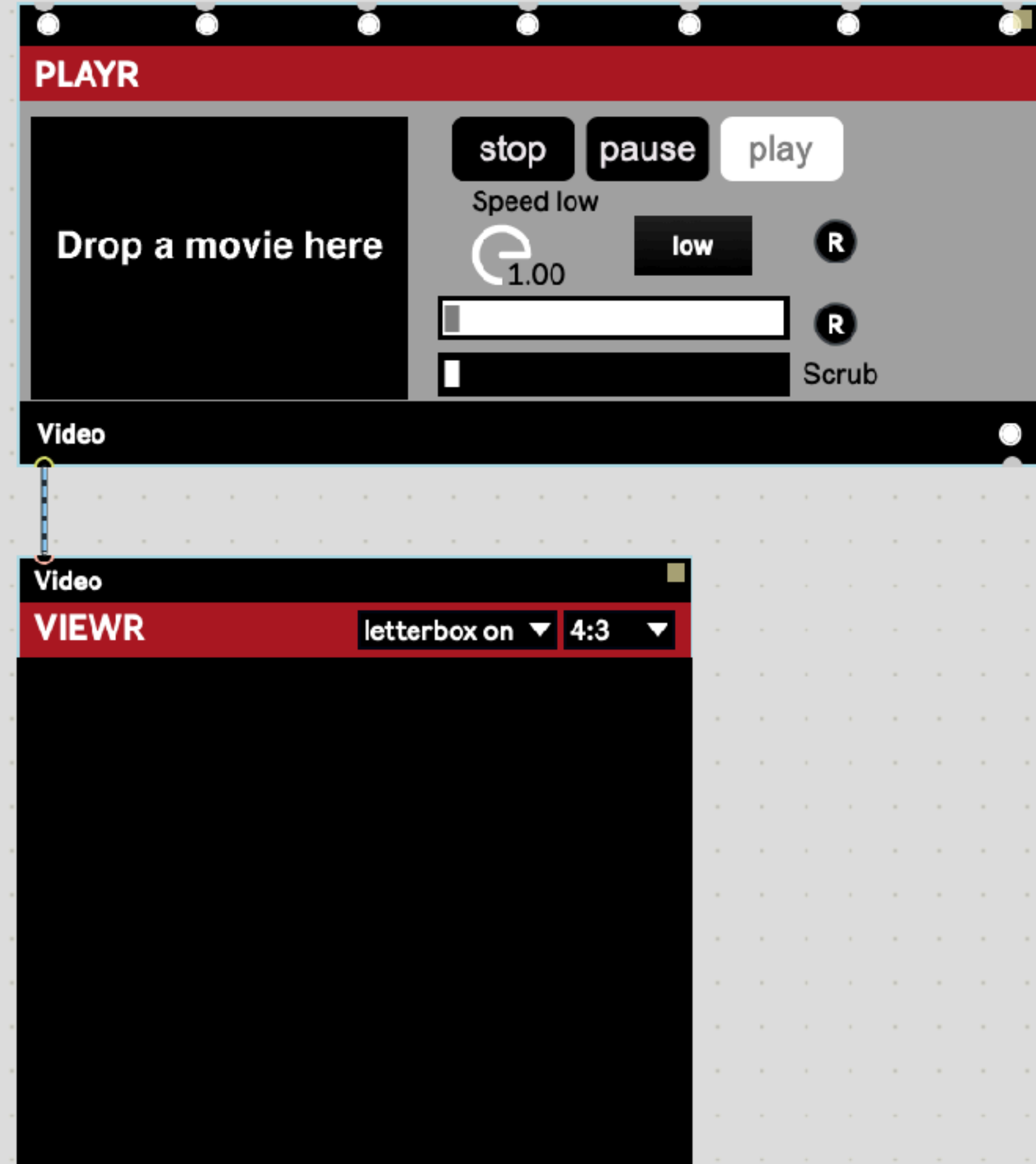


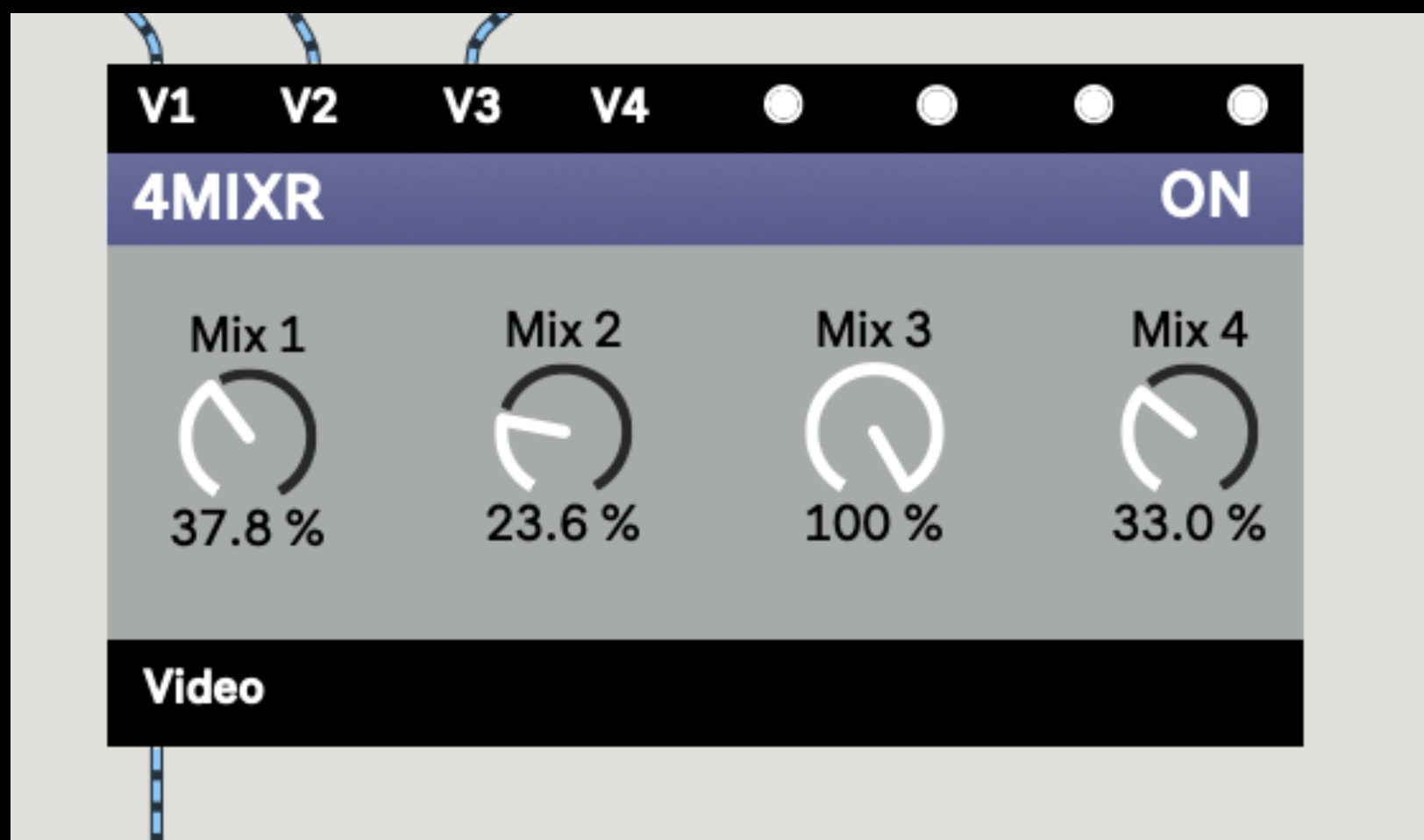
Video processing
patch

Vizzie is a collection of modules used to manipulate video and its very powerful! Find it in the lower left corner of your tool bar.

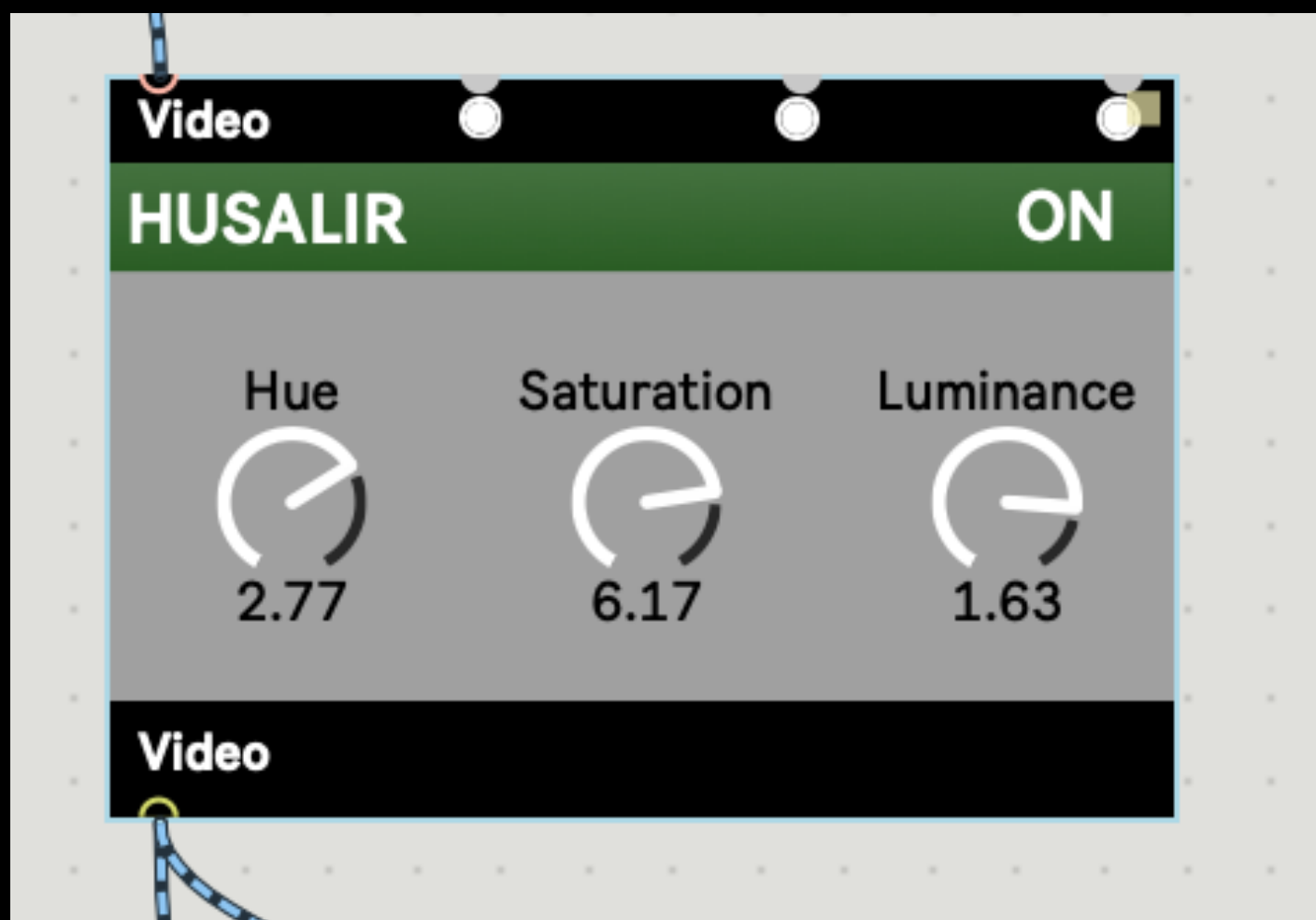


Vizzie models are just patches, but they are arranged in a way that makes them easy to use. Start by making an input (playr) and output (viewr)

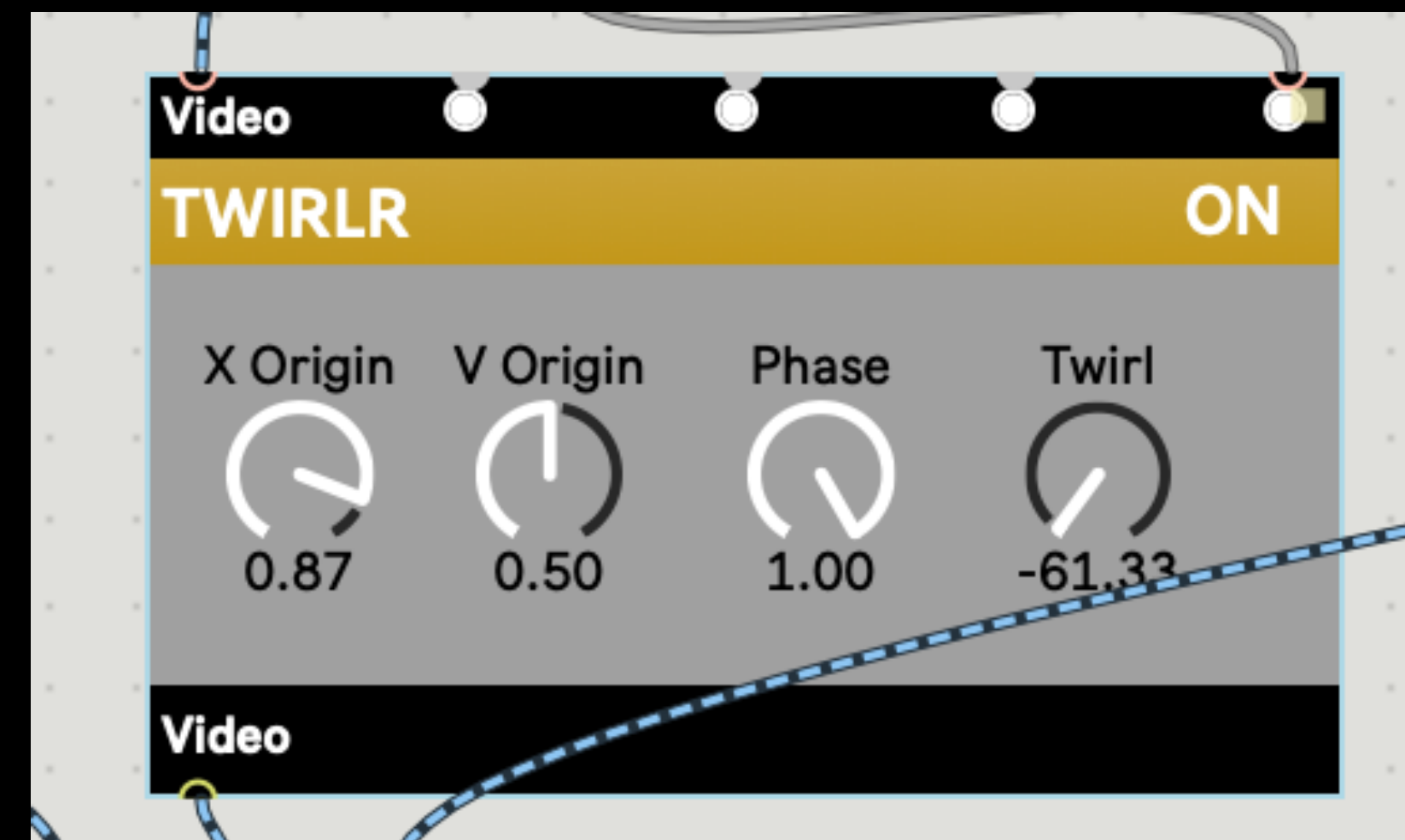




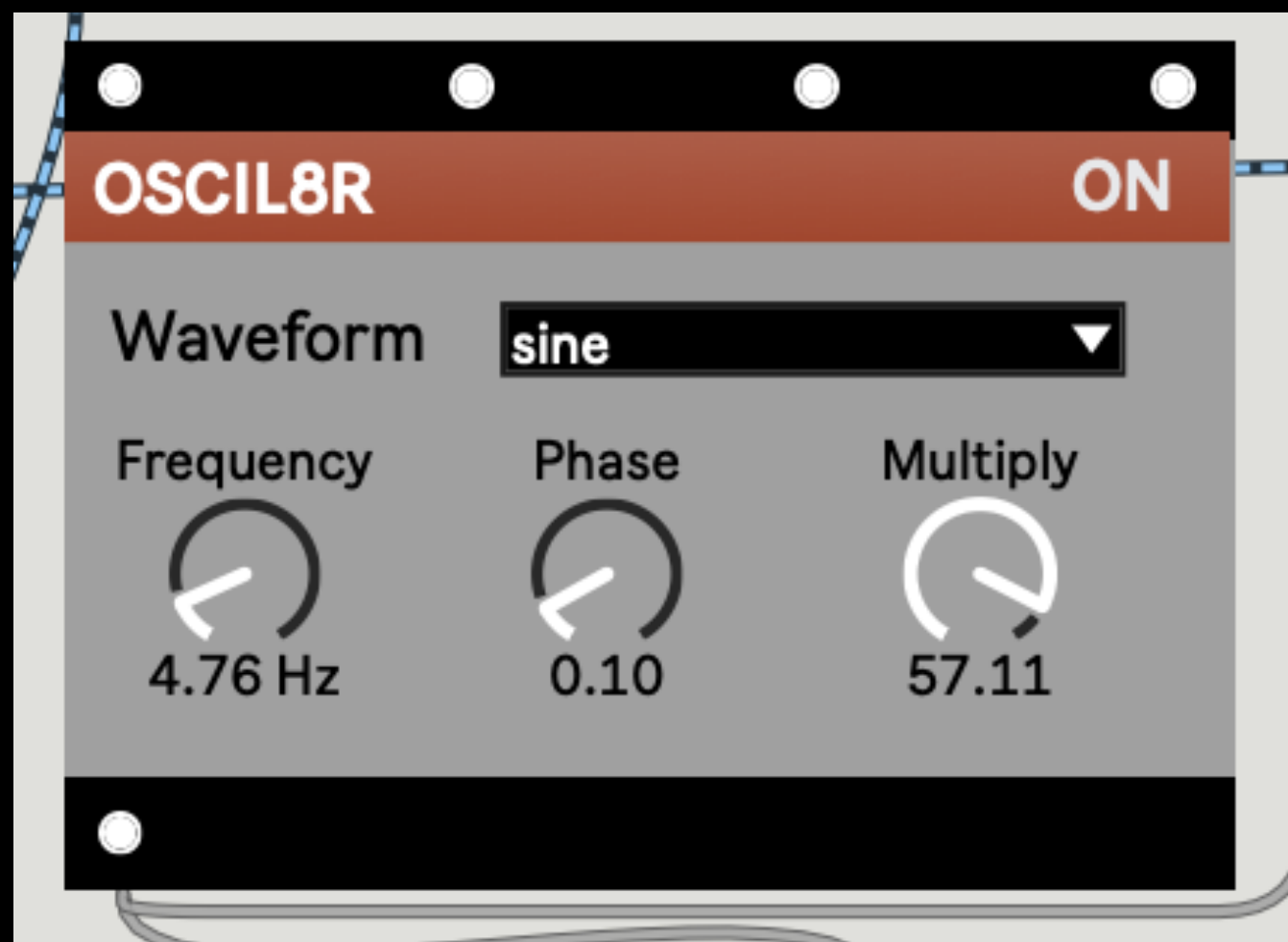
Mix



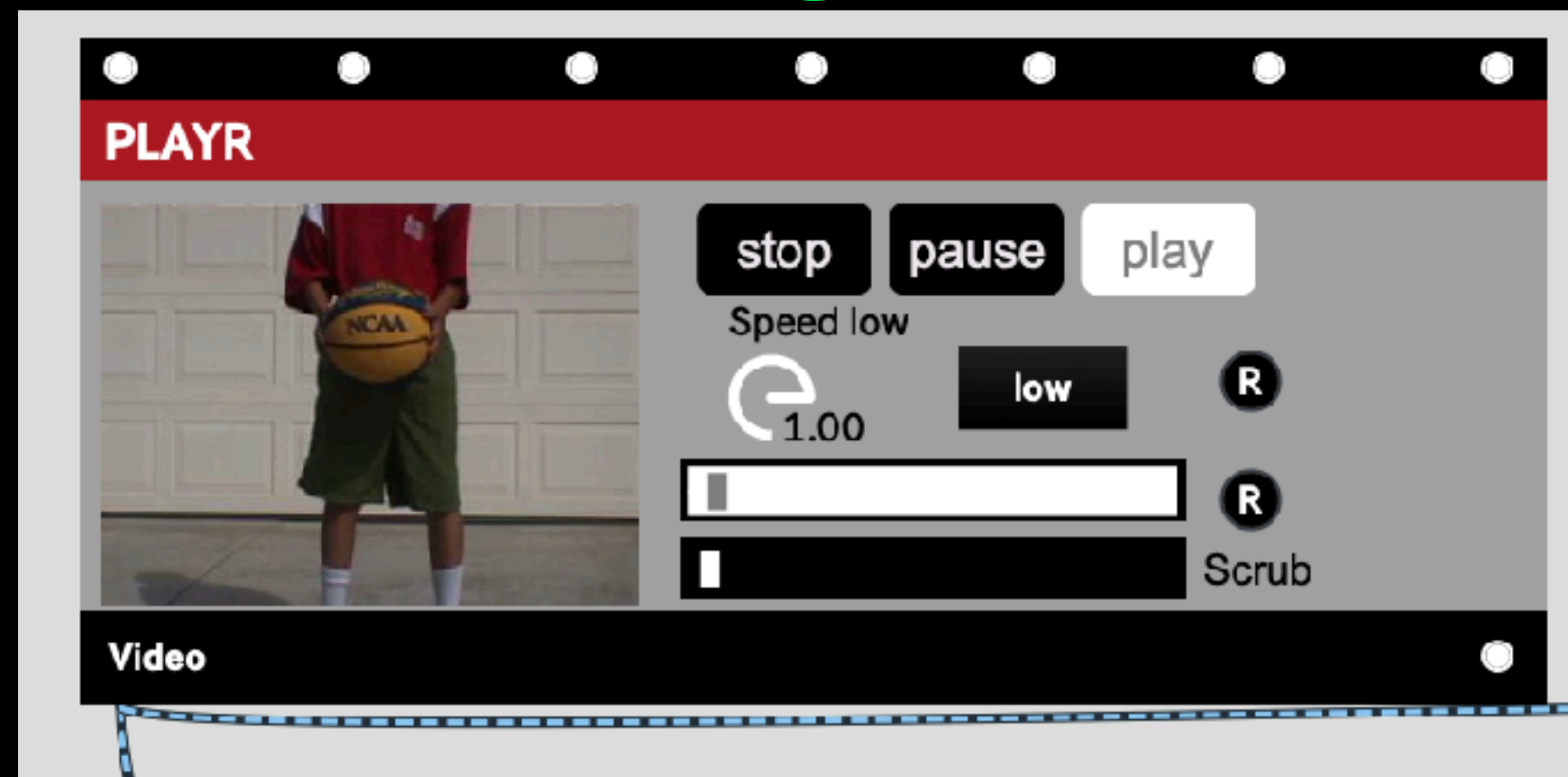
Change



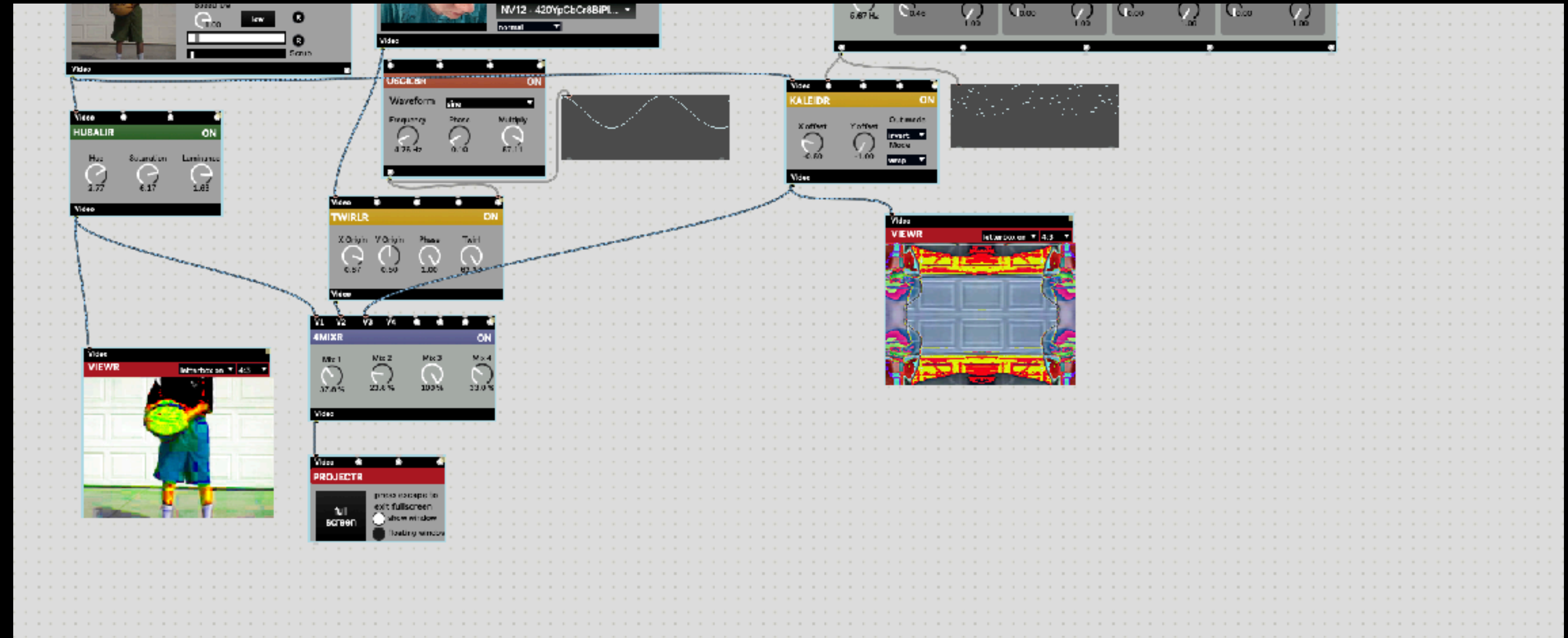
Effect



Generate



View



Vizzie Sample
patch, take some
time to play with
Vizzie;

**Download today's patch
on Moodle**

Midterm: Transform - Groups of 2-3, 25%, October 21

In this project you are invited to look beyond the computer screen and transform a space or atmosphere. Using primarily audio and video (and other if you want to) MaxMSP tools, bring us into a transformed within (or nearby) our classroom. This could mean changing the mood, forcing us to change perspectives, or imagine something new. Your project should **not** appear as a max patch, but as a piece for people to approach and engage with through a linear progression or interactive element. Your work should in some way have an arch or experience that builds or changes throughout the experience.

Be intentional. What do you want us to experience, and how are you bringing us there?

This work should feel complete - take note of where wires and cables are. Use projectors, lights, staging, curation etc to create a space. The work should be approaching gallery-ready state. Feel free to re-arrange any nearby space or reserve a critique room.

Homework:

Work on video Experiment (due next week)

Work on Midterm project

Next week: Using the camera, computer vision (open cv) and chromakey.